ENGINEERING REPORT — REV A · 20 SHEETS

Team Name

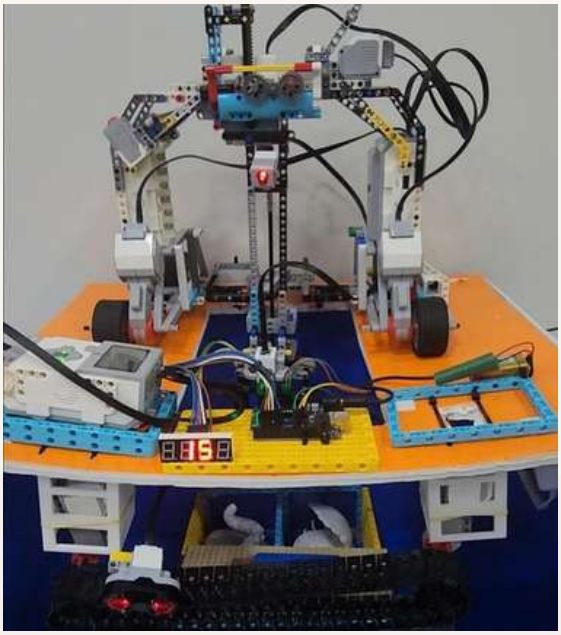
Aqsa Aqua Archaeo

IntelligentUnderwaterCulturalHeritage Explorer

A robot that scans the seabed, captures candidate images, and decides with **predefined safe actions** — across twenty sheets we documented every engineering decision: research, mechanics, embedded sensing, AI vision, and cultural-heritage impact, each backed by the team's evidence.

IDEA & RESEARCH | SYSTEM & MECHANICS | SOFTWARE & AI | IMPACT & INNOVATION | EVALUATION & ROADMAP | 20 SHEETS

"Discover the Past. Protect the Future."



01 idea&resarch

02 System & Mechanics P08 – P11

03 Software & AI

04 Impact & Innovation P15 – P17

05 Evaluation & Roadmap P18 – P19

CATEGORY	HARDWARE	SECTIONS / SHEETS	TEAM / SEASON
WRO — Future Innovators · Senior	4× EV3 + Arduino + Camera + AI vision	5 SECTIONS · 20 SHEETS	Aqsa Aqua Archaeo

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Team & Method

Members,coach,roles,andtheworking principle this report is built on

AQSA-Saudi-Stem is a student robotics team focused on applying engineering, programming, artificial intelligence, and STEAM thinking to real-world challenges. For WRO 2026, the team selected the protection and preservation of cultural heritage and developed NILEX as an autonomous/semi-autonomous underwater exploration concept.

Team Members

— click the spaces to add photos

Student 1 — Leen Adel

MECHANICAL · PROPULSION · TESTING

Main responsibilities: chassis, propulsion and arm design; waterproofing tests and structural stability measurements after every change.

Student 2 — Sara sultan

AI VISION · ARDUINO · SOFTWARE

Main responsibilities: AI pipeline and Arduino scanning subsystem; threshold tuning, camera positioning and test documentation.

COACH

Coach — Mohammed Basuny

Guidestheteam withoutdrivingit:asksthehardquestions, makes sure every engineering decision is truly ours, and holds us to documentation first.

Responsibilities — Engineering

- **Mechanical:** chassis, propulsion, arm mechanism, waterproofing concept, structural stability.
- **Robotics & Control:** EV3 architecture, motor control, sensor integration, autonomous routines.
- **Embedded:** Arduino scanning/sensing subsystem, communication interfaces, data acquisition.

Responsibilities — Research & Presentation

- **AI & Computer Vision:** image capture, preprocessing, artifact classification, confidence evaluation.
- **Research & Documentation:** heritage research, survey design, testing documentation, impact and entrepreneurship.
- **Presentation:** demonstration script, project booth, technical explanation and judge Q&A.

Team Working Principle

Research → Define → Design → Build → Program → Test → Measure → Improve

MAP

Document Map & Reading Guide

Where to find every requirement across the twenty sheets

The document is designed to be read as one continuous path: from problem to system to software to impact. Each of the five sections owns dedicated sheets flagged by its color, and every engineering claim is backed by the team's research.

Shee	Section	Group	Highlights
t	Cover	COVER	Team identity, scope, five section badges
P01	Team & Method	TEAM	Members, coach, roles, working principle
P02	Document Map	MAP	This map — full sheet tracking
P03	1.1 Problem & Research	SECTION 01	Problem statement, five research questions
P04	1.2 Design Requirements	SECTION 01	Eight requirements, research-to-design link
P05	1.3 Project Idea Summary	SECTION 01	Concept, AI decisions, non-invasive-first
P06	1.4 Idea Development	SECTION 01	Options A–D, why NILEX, engineering loop
P07	2.1 System Architecture	SECTION 02	Six layers: physical → control → AI → docs
P08	2.2 Four-EV3 Control	SECTION 02	EV3-1…EV3-4 modules, why four controllers
P09	2.3 Arduino Scanning	SECTION 02	Scan cycle, data exchange, calibration
P10	2.4 Mechanical Construction	SECTION 02	Frame, propulsion, arm rules, camera
P11	3.1 Vision & AI Pipeline	SECTION 03	Ten steps, thresholds, dataset strategy
P12	3.2 Autonomous Decisions	SECTION 03	State model, decision table
P13	3.3 Software Architecture	SECTION 03	Five modules, pseudo-code, safe AI mapping
P14	4.1 Social Impact	SECTION 04	Beneficiaries, impact, cultural principle
P15	4.2 Real-World Use Case	SECTION 04	Seven-phase submerged heritage survey
P16	4.3 Innovation & Responsibility	SECTION 04	Six innovations, business model, responsible AI
P17	5.1 Testing & Engineering	SECTION 05	Challenges, tests A–E, results table
P18	5.2 Next Steps & Metrics	SECTION 05	Roadmap short/medium/long, success metrics
P19	Closing & Sources	CLOSE	Conclusion, sources, revision log
P20			

Logical sequence

Each section builds on the previous: research shapes requirements, requirements shape the system.

Claim ≠ evidence

No judgment in this file without a measurement, a test or a defined rule behind it.

Photo spaces

Pale areas on each sheet are reserved for the team's real build and test photographs.

1.1 Problem, Research & Need

Whysubmergedculturalheritageishardtoexplore—andwhatwe asked

Problem Statement

Submerged cultural heritage can include shipwrecks, pottery, tools, sculptures, structures, and other objects that provide evidence about human history. Exploration may be complicated by poor visibility, depth, currents, unstable surfaces, and the need to protect fragile materials.

Exploration Constraints

DEPTH | LIMITED VISIBILITY | CURRENTS | DIFFICULT ACCESS | UNSTABLE SURFACES | RISK TO DIVERS | FRAGILE MATERIALS

Research Questions

1. How can underwater exploration become **safer and more repeatable**?
2. How can candidate objects be detected **without immediately disturbing the site**?
3. How can images be **documented and analyzed consistentl y**?
4. How can a robot **make decisions** from sensor and vision data?
5. How can the system **support archaeologists** rather than making unsupported archaeological claims?



The core constraint: the problem is not only finding objects — it is protecting fragile materials and the historical information they carry while exploration happens.

1.2 Design Requirements

Eight requirements that every subsystem must satisfy

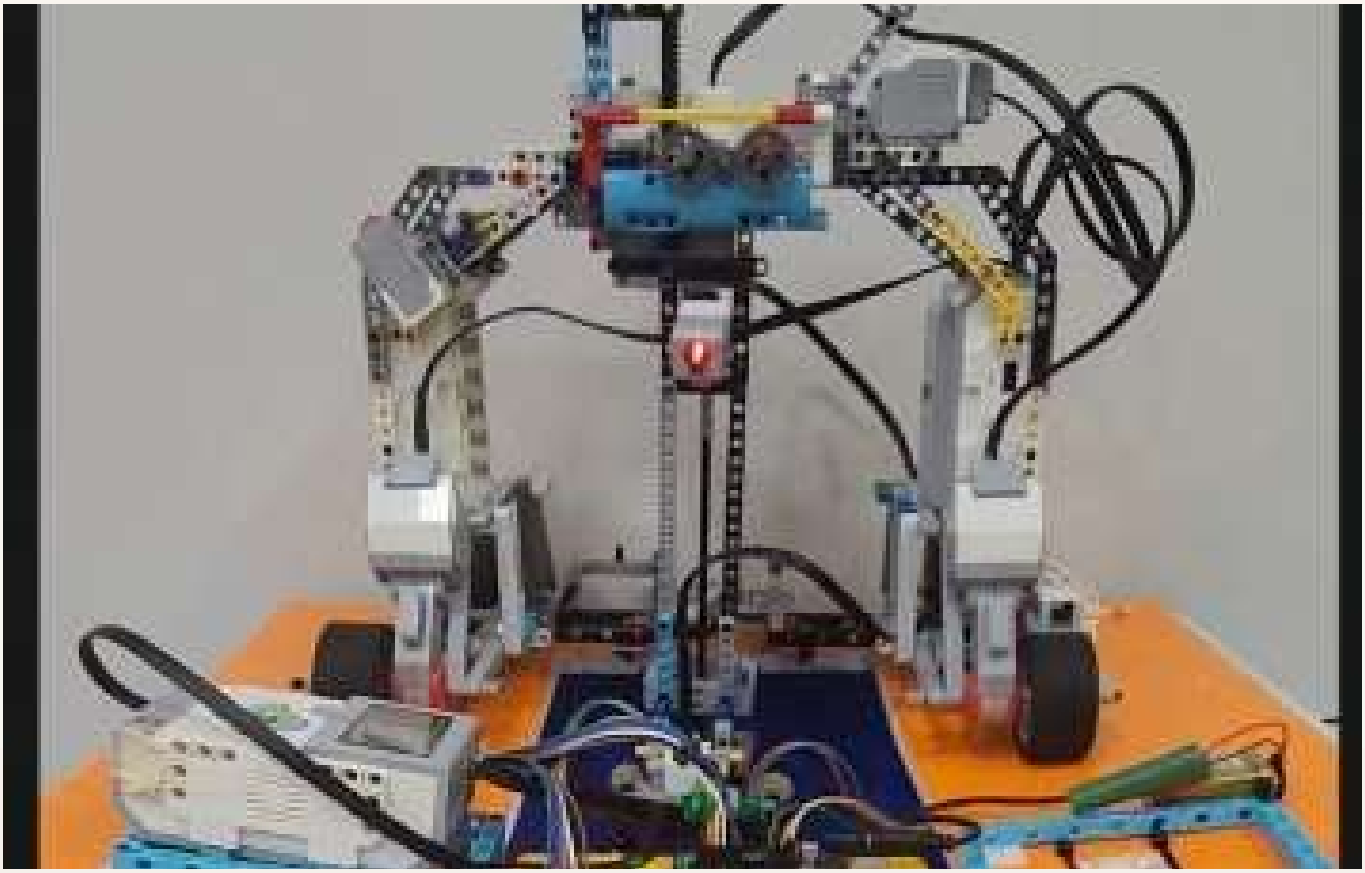
Design Requirements

- ✓ Scan a defined area systematically.
- ✓ Detect obstacles and potential objects.
- ✓ Capture usable images.
- ✓ Send image data to an AI analysis pipeline.
- ✓ Convert AI output into safe, predefined robot actions.
- ✓ Record the location and decision for each candidate.
- ✓ Operate with a non-invasive-first strategy.
- ✓ Provide human override and expert confirmation.

REQUIREMENTS	STRATEGY	AI OUTPUT	FINAL SAY
8	NON-INVASIVE	SAFE ACTIONS	HUMAN + EXPERT

Research-to-Design Link

The research led us to prioritize sensing, imaging, documentation, and decision support rather than simply building a robot that physically collects objects. This directly connects the engineering solution to the cultural-preservation problem.



Design consequence: every requirement above traces back to a research question — nothing in the architecture exists without a reason the team can defend.

1.3 Summary of the Project Idea

Oneworkflow:scan,detect,capture,analyze—andactonlythroughsafe rules

The Concept

NILEX is an intelligent robotic system designed to support the discovery, documentation, and protection of submerged cultural heritage. Our solution moves the first stage of exploration toward a robotic platform that can scan an area, detect objects of interest, capture images, and use artificial intelligence to assist in identifying possible archaeological artifacts.

The Prototype & Its Decisions

Four EV3 controllers coordinate movement, sensing, and mechanisms, while Arduino performs area scanning and environmental/object sensing. A camera captures candidate objects, and images are transmitted to an AI image-analysis pipeline. The AI does not generate uncontrolled robot code — it returns a structured decision, converted into predefined safe actions:

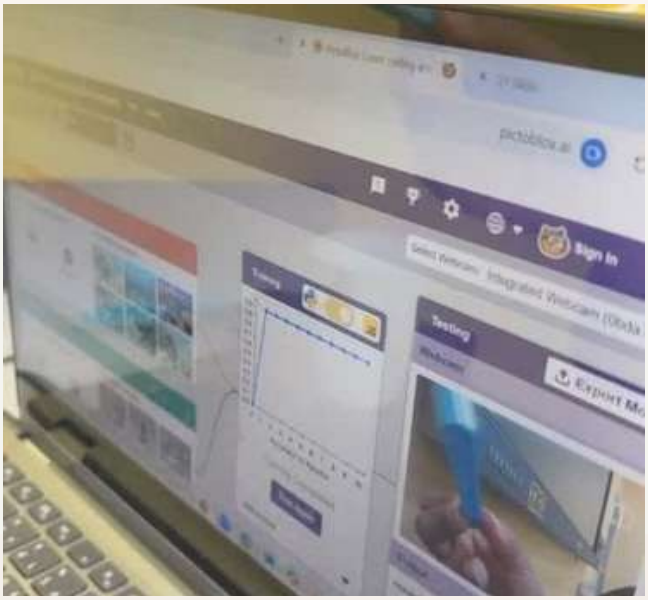
`SEARCH` `INSPECT` `CONFIRM` `AVOID` `CONTINUE`

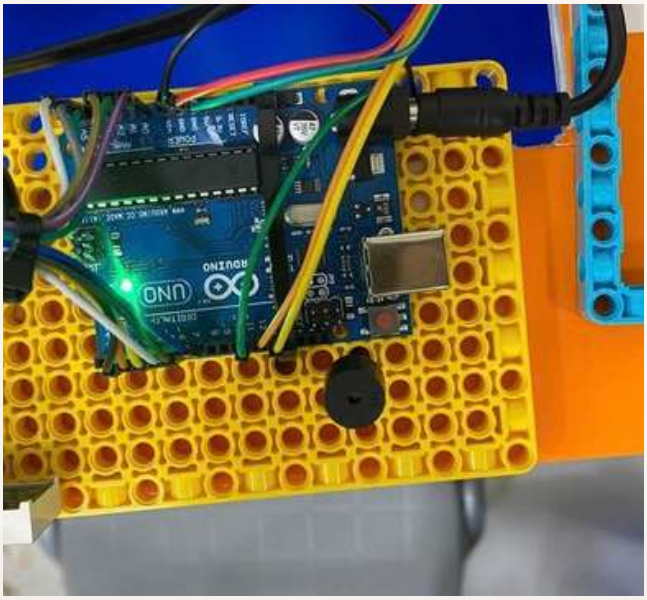
Non-Invasive by Design

NILEX is intended to observe and document first, rather than disturb a site. The robotic arm is used for controlled inspection only where appropriate, under defined safety rules.

The Value of NILEX

Robotics + embedded sensing + computer vision + AI decision support in one workflow — helping archaeologists survey difficult underwater areas, reducing human exposure, and creating a digital record of candidates.





Our goal is not to replace archaeologists. NILEX is a robotic assistant that collects evidence and supports expert decision-making.

1.4 Idea Development & Differentiation

Options investigated, why NILEX, and what makes our approach different

A
A manual remotely operated vehicle controlled entirely by a human.

B
A camera-only underwater inspection robot.

C
A sensor-based autonomous exploration robot.

D
An integrated robotic system combining scanning, imaging, AI analysis, and autonomous decision support.
— chosen

Why NILEX?

A camera-only system cannot connect images to movement and sensing decisions; a manual vehicle depends on operator attention; a sensor-only system cannot provide visual evidence. NILEX combines these functions into a closed engineering loop:

`Sense` → `Move` → `Detect` → `Capture` → `Analyze` → `Decide` → `Act` → `Document`

What Is Different About Our Approach?

1. Multi-controller architecture using four EV3 units.
2. Arduino-based scanning subsystem for dedicated processing.
3. AI-assisted image analysis linked to predefined behaviors.
4. Non-invasive-first exploration and documentation.
5. A structured decision interface between AI and robot control.
6. Modular design so future teams can replace subsystems.



The innovation is therefore not a single component. It is the integration of several technologies into a workflow designed specifically for cultural-heritage protection.

2.1 System Architecture — Six Layers

From the physical platform to documentation, one layer at a time

Layer 1 — Physical Platform

CHASSIS · PROPULSION · ARM · CAMERA MOUNT · SENSOR MOUNTS · LIGHTING

Layer 2 — Robotic Control

EV3-1: locomotion & navigation
EV3-2: environmental & obstacle sensing
EV3-3: inspection/arm mechanisms
EV3-4: coordination, logging, redundancy

Layer 3 — Arduino Scanning

Dedicated scanning subsystem sweeps the target area and identifies regions requiring closer inspection.

Layer 4 — Vision & AI

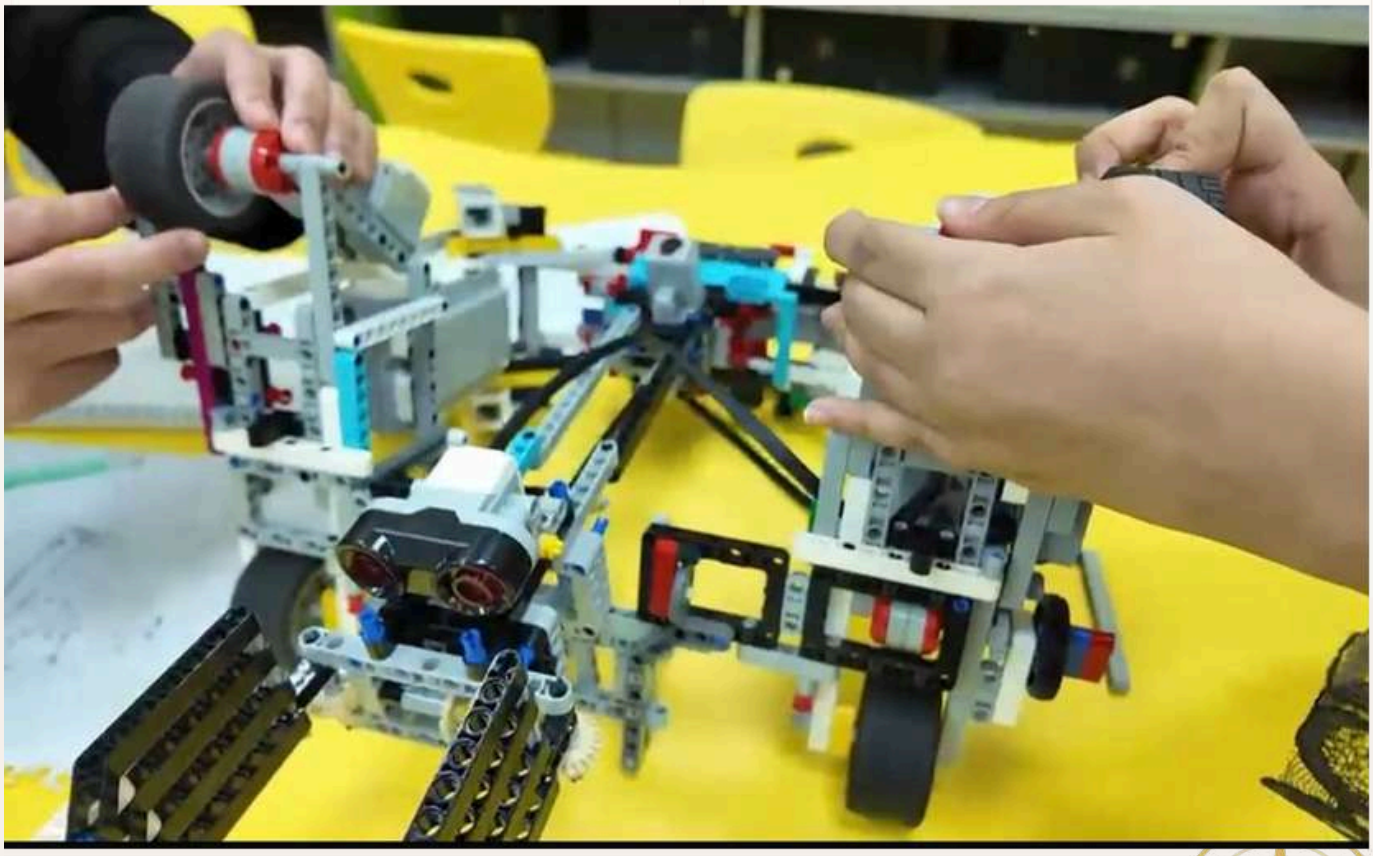
Camera → Capture → Preprocess → Classify → Decision

Layer 5 — Decision & Safety

SEARCH · INSPECT · CONFIRM · AVOID · CONTINUE · RETURN

Layer 6 — Documentation

TIMESTAMP · ZONE · IMAGE · AI RESULT · CONFIDENCE · ACTION



Safety by separation: this architecture separates perception from actuation — reducing the risk of an AI classification error directly causing an uncontrolled physical action.

2.2 Four-EV3 Control Architecture

Independent functional modules — debug, expand, and trust each one

EV3-1 — Navigation

- **Controls:** left/right propulsion · forward/reverse · tur ning
- Encoder-based distance control
- Navigation states

EV3-2 — Sensing

- **Processes:** distance/obstacle sensors
- Environmental readings
- Safety boundaries · scan-zone info

EV3-3 — Inspection

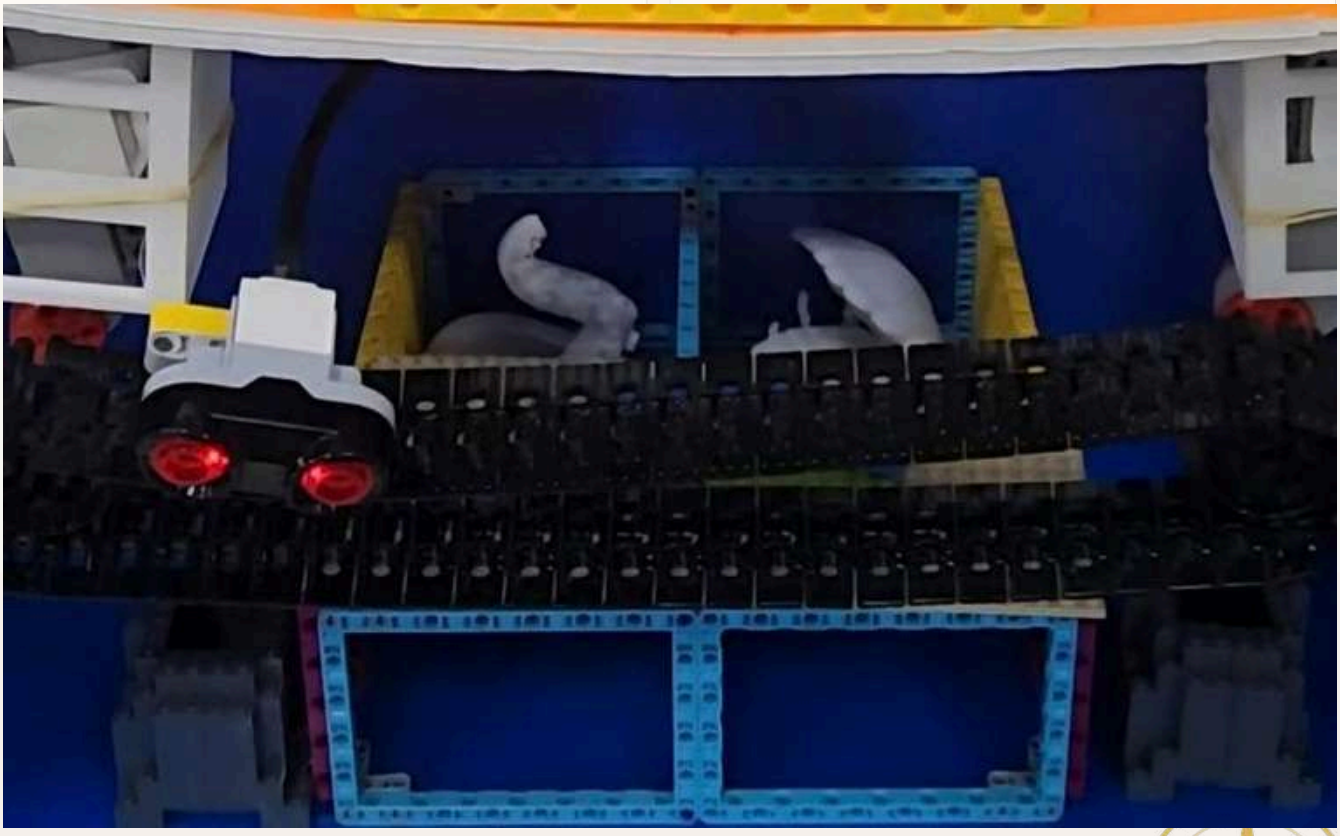
- **Controls:** robotic arm · camera positioning
- Inspection movement
- Controlled approach to candidates

EV3-4 — Coordination

- **Manages:** mission states · communication
- Event logging · recovery routines
- Human override interface

Why Four Controllers?

A modular architecture makes debugging easier. If one subsystem fails, the team can isolate the fault instead of troubleshooting the entire program as one large routine. It also supports future expansion: additional sensors or an improved AI interface can be added without rewriting every movement routine.



Important design principle: the AI layer provides decisions; the EV3 controllers execute only predefined, tested actions.

2.3 Arduino Scanning Subsystem

A dedicated embedded unit for high-frequency sensing

Scanning Objective & Cycle

The Arduino subsystem sweeps the target area, processes readings locally, and passes a compact result to the robot-control layer:

1. Start scan.
2. Read sensor array.
3. Compare with thresholds.
4. Estimate object/obstacle region.
5. Assign scan-zone identifier.
6. Request camera inspection.
7. Return result to coordination layer.

Data Exchanged

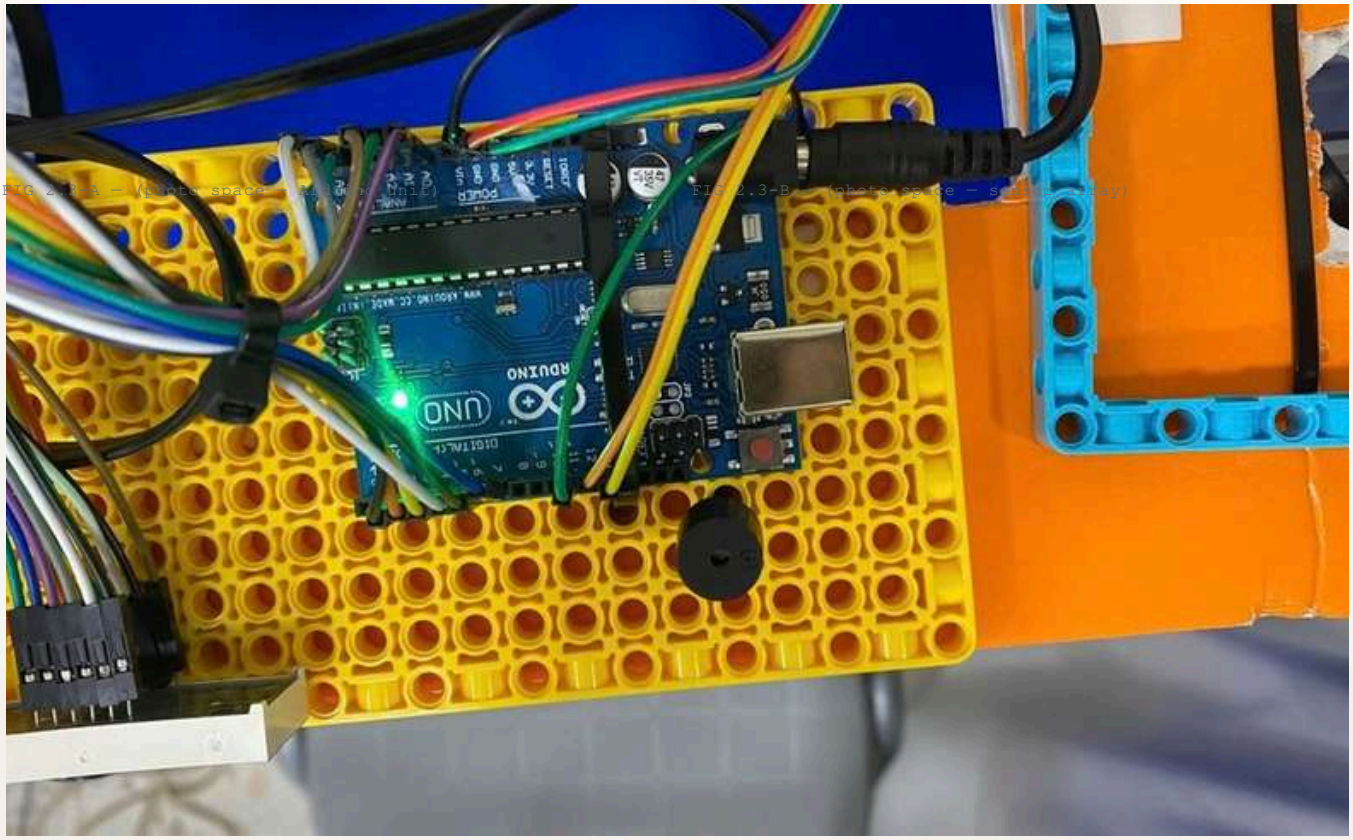
- SCAN ZONE
- SENSOR VALUES
- OBJECT DISTANCE
- DETECTION FLAG
- SAFETY FLAG
- TIMESTAMP

Why Arduino?

A dedicated embedded controller allows high-frequency sensor processing without placing all sensing tasks on one EV3 controller.

Engineering Benefit

The subsystem is modular. The same concept could later support additional sensors such as depth, temperature, conductivity, or other underwater sensing technologies, depending on the final hardware and scientific requirements.



Calibration rule: the prototype will be calibrated using known objects and controlled test conditions before any field-oriented demonstration.

2.4 Mechanical Construction

Modulardesignfocusedonstability,visibility,protection,and serviceability

Main Structure & Propulsion

- Rigid central frame · distributed mass
- Protected electronics compartment concept
- Modular sensor brackets · adjustable camera mount
- Front inspection/arm module

Propulsion is controlled by EV3 motor groups; closed-loop control with encoder feedback maintains consistent, repeatable movement during scanning.

Robotic Arm, Camera & Lighting

The arm is designed for controlled inspection. The preferred operating rule is:

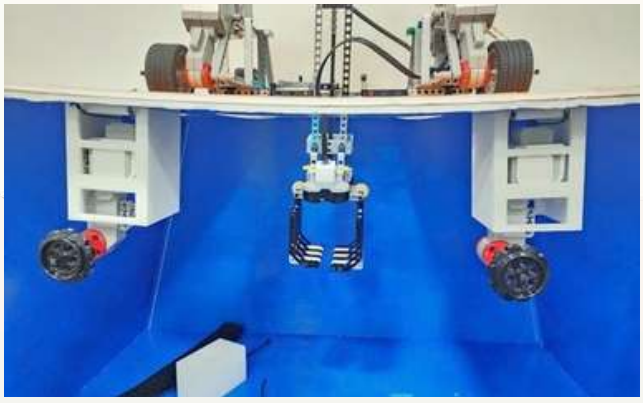
OBSERVE → CLASSIFY → CONFIRM → INTERACT

The arm is not intended for uncontrolled excavation — any physical interaction would require conservation protocols and expert authorization. Lighting reduces shadows and improves image consistency.

Engineering Considerations

CENTER OF MASS · FRAME STIFFNESS · CABLE ROUTING · FIELD OF VIEW · MAINTENANCE ACCESS · WATER-RESISTANT ENCLOSURE · REPEATABLE CAMERA POSITION





Next validation step: future prototype work will validate buoyancy, sealing, propulsion efficiency, and underwater stability using an appropriate test environment.

3.1 Computer Vision & AI Pipeline

Assist — not replace — archaeological expertise

Pipeline — Ten Steps

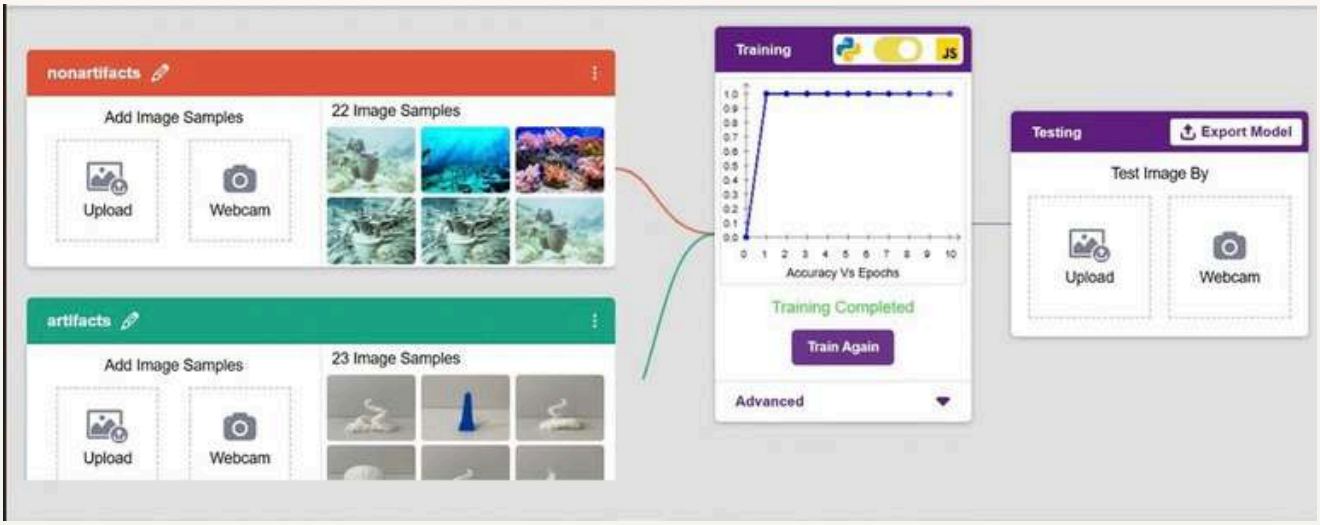
01 Camera captures candidate image.
02 Image quality check.
03 Crop and normalize the region.
04 Remove irrelevant background.
05 Enhance contrast/visibility.
06 AI model evaluates the image.
07 Model returns class + confidence.
08 Decision layer applies thresholds.
09 Robot receives a safe command.
10 Image and decision are logged.

Example Output

Class: `POTENTIAL_ARTIFACT` · Confidence: `0.87` · Decision: `INSPECT`
Below threshold → **CONTINUE_SCAN** · Obstacle → `AVOID`

AI Training Strategy

`ARTIFACT-LIKE SAMPLES` `ROCKS` `DEBRIS` `SAND` `LOW-LIGHT` `BLURRED` `ORIENTATIONS`



Honest metrics & human-in-the-loop: we will not claim an accuracy value that has not been experimentally measured — and a high-confidence AI result is still a candidate classification. Archaeological confirmation remains the responsibility of qualified experts.

3.2 Autonomous Decision-Making

A finite-state decision model — explainable and testable

States 1–5

STATE 1 — SEARCH: move through the defined scan path.
STATE 2 — DETECT: evaluate sensor readings; identify a candidate region.

STATE 3 — INSPECT: slow down, position camera, capture images.

STATE 4 — ANALYZE: send the image through the AI pipeline.

STATE 5 — DECIDE: convert AI output into a predefined safe command.

States 6–7

STATE6A—CONTINUE: no relevant candidate: resume scanning.

STATE 6B — AVOID: obstacle detected: change path safely.

STATE 6C — DOCUMENT: record image, position, confidence, time.

STATE 6D — HUMAN CONFIRMATION: require authorization before physical interaction.

STATE 7 — RETURN: complete the mission or return to start.

Obstacle	`AVOID`	Potential artifact	`DOCUMENT + INSPECT`
Low image quality	`RECAPTURE`	High-confidence candidate	`DOCUMENT + REQUEST_CONFIRMATION`
Low confidence	`CONTINUE_SCAN`	Mission complete	`RETURN`

Why this works: this approach makes the autonomous behavior explainable and testable.

3.3 Programming & Software Architecture

Independent modules and a main loop that never acts on unverified AI output

Navigation
Motor control · encoder feedback · speed profiles · turning routines

Sensing
Sensor reading · filtering · threshold detection · safety checks

Vision
Image capture · preprocessing · AI request/response handling

Decision
Confidence thresholds · state transitions · safety rules · override

Logging
Scan zone · detection event · image reference · AI result · action

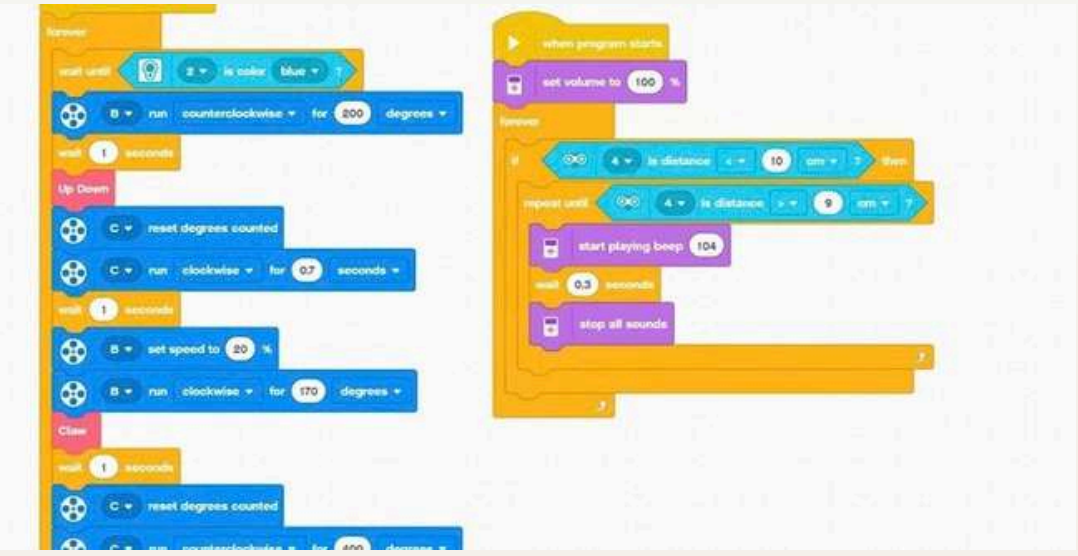
Pseudo-Code

```
START
Initialize EV3 controllers
Initialize Arduino scanning
Check safety status

WHILE mission_active:
  scan_area()
  IF obstacle_detected:
    avoid_obstacle()
  ELSE IF candidate_detected:
    capture_image()
    result = AI_ANALYZE(image)
    IF result == "LOW_CONFIDENCE":
      continue_scan()
    ELSE IF result == "POTENTIAL_ARTIFACT":
      document_candidate()
      move_to_inspection_position()
    ELSE:
      continue_scan()
  log_event()

IF mission_complete:
  return_to_start()
END
```





Final implementation: will use tested block-based and/or embedded code appropriate to the selected hardware. AI output will map to predefined commands rather than arbitrary generated code.

4.1

Social Impact & Beneficiaries

Supportingthe people whoprotect andstudy culturalheritage

PRIMARY GROUPS	SECONDARY GROUPS	RECORD	FIRST RULE
5	5	DIGITAL	NON-INVASIVE

Primary Beneficiaries

- Archaeologists
- Cultural-heritage authorities
- Conservation specialists
- Museums and research institutions
- Marine survey teams

Secondary Beneficiaries

- Students and educators
- Museums and visitors
- Researchers
- Local communities connected to heritage sites
- Future generations

Potential Impact

NILEX can reduce the need for people to enter hazardous or difficult environments during the first stage of exploration. It can also create a repeatable digital record of candidate locations and images. The social value is not only finding an object — it is protecting information about the object and its context.

نتائج استبانة الطلاب		
أولاً: النتائج الرئيسية		
البند	النتيجة	النسبة
الوظيفة الأكثر أهمية	الوصول إلى الآثار في المواقع الصعبة والمتردية	45%
التقنية الأكثر أهمية	دمج جميع التقنيات (حساسات + كاميرات + تحليل صور)	55%
التحدي الأكبر لاكتشاف الآثار	صعوبة الوصول وضعف الرؤية تحت الماء	75%
ثانياً: نتائج الاستبيان		
السؤال	توزيع الإجابات	نسبة الموافقة
هل يشكّل الروبوت تحدياً لأدوات الاكتشاف التقليدية؟	أوافق بشدة 60% • أوافق 30% • محايد 5% • لا أوافق 5%	90%
ما مدى أهمية فهم الروبوت للذكاء الاصطناعي؟	أوافق بشدة 45% • أوافق 30% • محايد 15% • لا أوافق 10%	75%
هل تغطي الروبوتات أغلب المهام المطلوبة؟	أوافق بشدة 75% • أوافق 20% • لا أوافق 5%	95%

Cultural principle: a non-invasive-first philosophy — observe, document, analyze, and preserve. Any physical intervention should follow expert conservation requirements.

4.2 Real-World Use Case

Submerged Heritage Survey — a seven-phase workflow

Example scenario: a cultural-heritage team identifies an underwater area that may contain historical remains.

Phase 1 — Mapping

NILEX scans the defined area and records movement and sensor information.

Phase 2 — Candidate Detection

The scanning system identifies an unusual object or region.

Phase 3 — Visual Inspection

NILEX positions its camera and captures several images.

Phase 4 — AI Assistance

The AI improves image quality and evaluates whether the pattern resembles a potential artifact.

Phase 5 — Decision

If confidence is insufficient, scanning continues; if relevant, the robot documents and requests the next authorized action.

Phase 6 — Expert Review

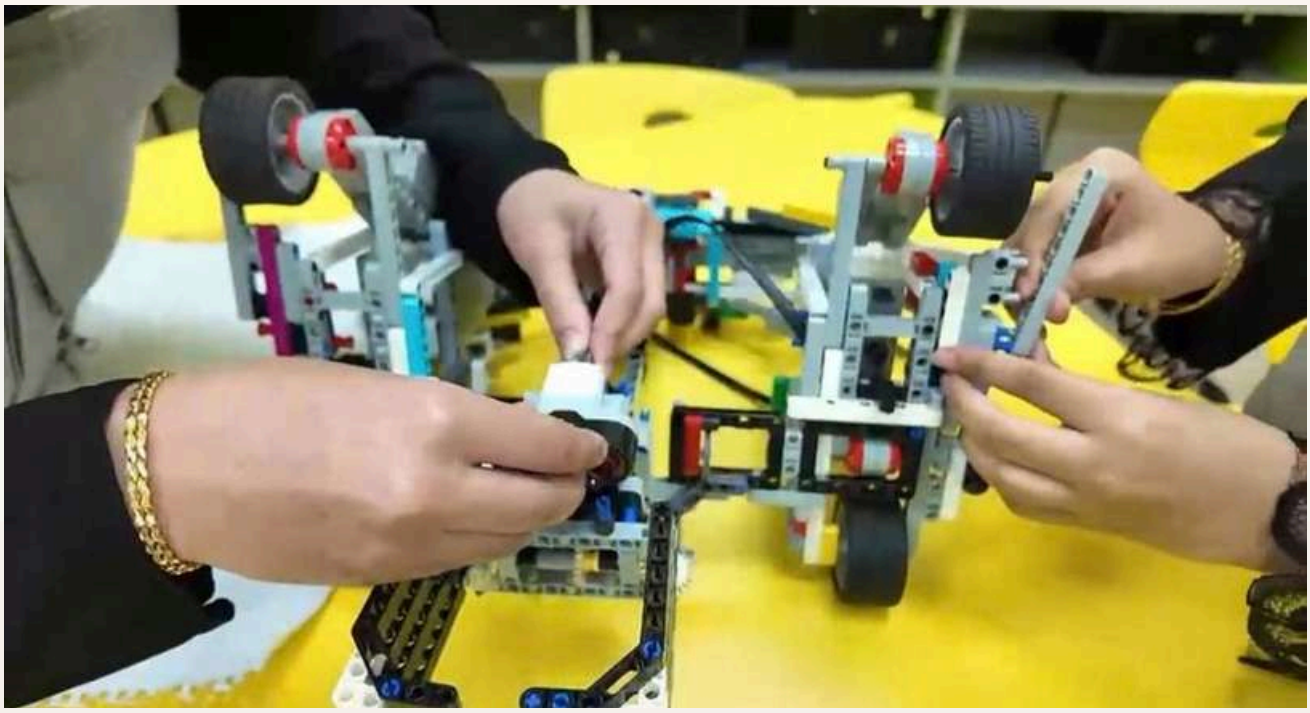
An archaeologist reviews the evidence and determines whether further investigation is justified.

Phase 7 — Digital Record

The final record can include location, images, classification, and inspection notes.

Result

The robot becomes a **field assistant** — evidence in, expert decision out.



Positioning: this workflow makes the robot a field assistant rather than an autonomous archaeological authority.

4.3 Key Innovation, Business & Responsible AI

Integration of autonomous robotics and AI-assisted visual decision support

1 · Multi-controller robotics

Four EV3 controllers divide complex functions into manageable modules.

2 · Dedicated scanning

Arduino handles the scanning subsystem so sensing operates independently.

3 · AI-to-action interface

AI results convert into a small set of predefined, explainable commands.

4 · Evidence-first workflow

Every candidate is documented with image and context before interaction.

5 · Modular development

The architecture allows future improvements to sensors, cameras, models, software.

6 · Human-centered autonomy

Autonomy handles repetitive decisions; experts confirm interpretation and intervention.

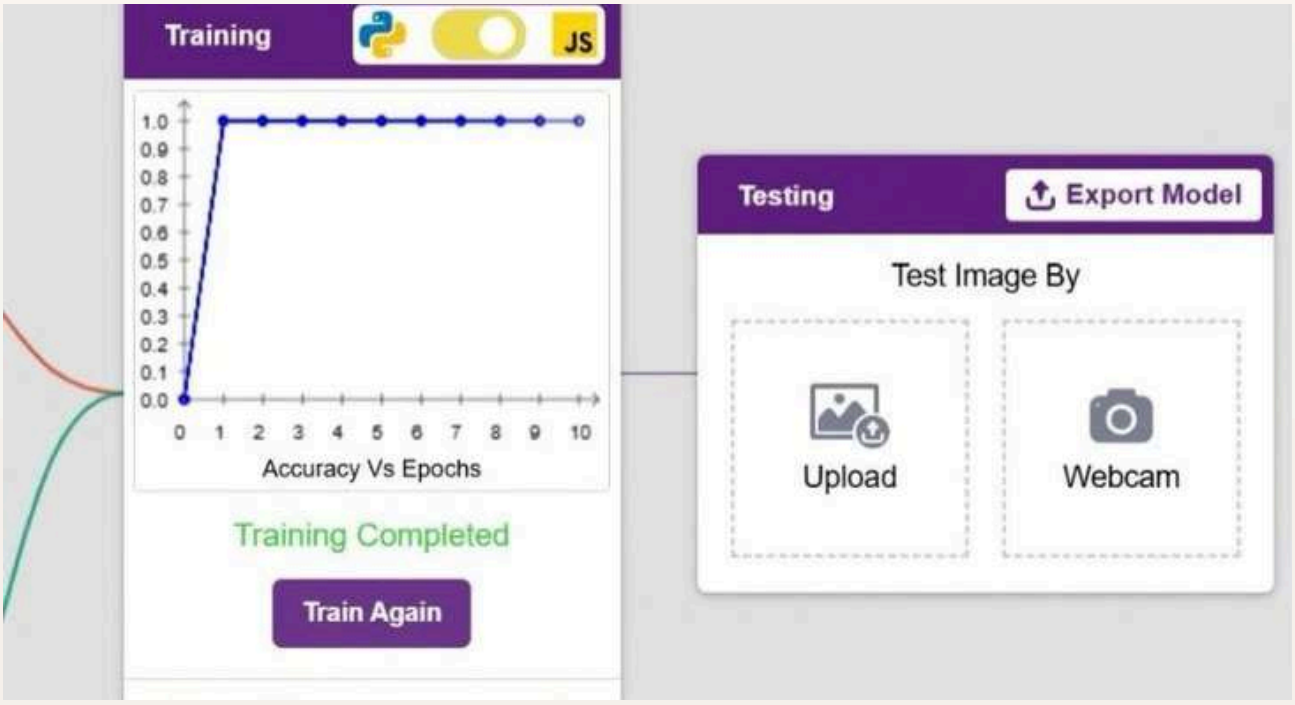
Entrepreneurship & Business Model

Users: heritage authorities · research institutions · museums · universities · marine survey organizations. **Value proposition:** a modular robotic assistant for safer surveying, documentation, detection, and AI-assisted inspection.

RESEARCH SERVICES | INSTITUTIONAL PILOTS | EDUCATIONAL VERSIONS | CUSTOM SYSTEMS

Sustainability, Accessibility & Responsible AI

Modular platform — repaired/upgraded, not replaced.
AI output is decision support; thresholds are explicit.
Low-confidence cases are not forced into a classification.
Human expertise stays in the final archaeological decision.
Remote expert review via transmitted images and structured info.
Heritage data handled per authorities' rules; locations never disclosed without authorization.



Key message: "Autonomy should increase exploration capability without reducing human responsibility."

5.1 Development Challenges, Testing & Engineering

One subsystem at a time — then integrate

Major Development Challenges

1. Coordinating multiple controllers.
2. Maintaining stable movement while scanning.
3. Synchronizing camera capture with robot position.
4. Obtaining consistent images under changing light.

...continued

5. Reducing false detections.
6. Designing a safe AI-to-robot interface.
7. Protecting electronics in the prototype environment.

Testing method: one subsystem at a time, then integrate.

Test Plan A–E

- **Test A — Navigation:** distance error and repeatability.
- **Test B — Scanning:** detection of known objects and false positives.
- **Test C — Vision:** image quality under different lighting and distances.
- **Test D — AI:** accuracy, precision, recall, and confidence behavior on a documented test set.
- **Test E — Integrated Mission:** successful completion of the complete sequence.

FIG 5.1-A — (photo space — test setup)

FIG 5.1-B — (photo space — test run)

Navigation repeatability	Define before testing	To be measured
Candidate detection rate	Define test set	To be measured
False-positive rate	Define test set	To be measured
AI accuracy	Use held-out test set	To be measured
Mission completion	Repeat integrated test	To be measured

Before submission: replace every “To be measured” field with verified experimental data.

5.2 Next Steps & Success Metrics

Roadmapandthenumbersthatwilldefine success

Short-Term

1. Complete integrated EV3/Arduino communication architecture.
2. Calibrate all sensors.
3. Improve camera positioning and lighting.
4. Build a controlled underwater test environment.
5. Create and label a project-specific image dataset.
6. Train and validate the AI model.
7. Measure false positives and negatives.
8. Improve navigation repeatability.

Medium-Term

Add depth sensing.
Improve underwater localization.
Add structured mission logs.
More robust waterproof enclosure.
Improve remote monitoring.
Test multiple object types and conditions.

Long-Term Vision

3D mapping
Higher-resolution imaging
Sonar or acoustic sensing
Improved localization
Expert collaboration Conservation-aware manipulation Digital heritage visualization

Success Metrics

Technical:

NAVIGATION REPEATABILITY | SCAN COVERAGE | DETECTION RATE | IMAGE QUALITY | AI PERFORMANCE | FALSE-POSITIVE RATE | MISSION COMPLETION | RESPONSE TIME

Social & Innovation:

REDUCED HUMAN EXPOSURE | DOCUMENTATION QUALITY | EXPERT USEFULNESS | INSTITUTIONS BENEFITING | INTEGRATION | MODULARITY | EXPLAINABLE DECISIONS | HUMAN-IN-THE-LOOP SAFETY



Prototype philosophy: each new feature must be tested independently, measured, and integrated only after the team understands its limitations.

Discover carefully. Document accurately. Protect responsibly.

EVIDENCE-FIRST | NON-INVASIVE-FIRST | HUMAN-IN-THE-LOOP | MODULAR | EXPLAINABLE

Conclusion

NILEX is a student-developed robotics concept that connects cultural heritage, robotics, embedded systems, computer vision, and artificial intelligence. The project responds to a real challenge: how to explore difficult underwater environments while preserving the evidence and context of cultural heritage. The system uses four EV3 controllers and an Arduino scanning subsystem to divide the engineering problem into modular components. A camera and AI pipeline provide visual decision support, while predefined control logic converts AI results into safe robotic actions.



List of Sources

- WorldRobotOlympiad Association. WRO 2026 – Future Innovators Mission: "Robots Meet Culture". Official mission document.
- WRO. WRO 2026 – Future Innovators Information for Teams & Coaches. Official project-report template and competition guidance.
- WRO. WRO 2026 Season page. Official category and season infor mation.
- WRO. Future Innovators category information and general rules.
- UNESCO. Underwater Cultural Heritage: official information and international framework for the protection of underwater cultural heritage.
- UNESCO 2001 Convention on the Protection of the Underwater Cultural Heritage.
- Team-generated survey: "Survey on Using Robots to Discover and Protect Submerged Cultural Heritage." Results to be inserted after data collection.
- Team engineering documentation: NILEX prototype design, testing records, code versions, sensor calibration logs and AI validation results.

Revision	Date	Change description
REV A	2026-00-00	First complete release of the unified engineering report — 20 sheets.
REV B	N/A	N/A

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DOCUMENT
WRO-FI-2026 · REV A

Before final WRO submission: replace all planned/target metrics with measured experimental results, add the final robot photographs, insert the actual survey statistics, and document the final hardware and software configuration. Verify the latest international rules and any national-organizer adaptations.